

Assignment 3

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1 Exercise 1.13

Consider the bin model for a hypothesis h that makes an error with probability μ in approximating a deterministic target function f (both h and f are binary functions). If we use the same h to approximate a noisy version of f given by:

$$P(y|x) = \begin{cases} \lambda & \text{if } y = f(x), \\ 1 - \lambda & \text{if } y \neq f(x). \end{cases}$$

- (a) What is the probability of error that h makes in approximating y ?
- (b) At what value of λ will the performance of h be independent of μ ?
[Hint: The noisy target will look completely random.]

Solution

(a) Probability of Error

We need to find the probability of error that h makes in approximating y .

There are two cases to consider:

- **Case 1:** $f(x) = h(x)$.
 - The probability that h is correct with respect to f is $1 - \mu$.
 - Given $f(x) = h(x)$, the probability that $y \neq f(x)$ (i.e., the noisy function gives an incorrect value) is $1 - \lambda$.

Therefore, in this case, the probability of error is:

$$(1 - \mu)(1 - \lambda)$$

- **Case 2:** $f(x) \neq h(x)$.
 - The probability that h makes an error with respect to f is μ .
 - Given $f(x) \neq h(x)$, the probability that $y = f(x)$ (the noisy function gives the correct value) is λ .

Therefore, in this case, the probability of error is:

$$\mu\lambda$$

The total probability of error that h makes in approximating y is the sum of the two cases:

$$P_{\text{error}} = (1 - \mu)(1 - \lambda) + \mu\lambda$$

(b) Value of λ where performance is independent of μ

To find the value of λ where the performance of h is independent of μ , we analyze the expression for the error probability:

$$P_{\text{error}} = (1 - \mu)(1 - \lambda) + \mu\lambda$$

We require that the derivative of P_{error} with respect to μ is zero. Differentiating with respect to μ :

$$\frac{d}{d\mu} [(1 - \mu)(1 - \lambda) + \mu\lambda] = -(1 - \lambda) + \lambda = 0$$

Solving this equation gives:

$$\lambda = \frac{1}{2}$$

The performance of h is independent of μ when $\lambda = \frac{1}{2}$.

2 Exercise 2.1

By inspection, find a break point k for each hypothesis set in Example 2.2 (if there is one). Verify that $m_{\mathcal{H}}(k) < 2^k$ using the formulas derived in that Example.

Break Point k

In Example 2.2, we examine the hypothesis set of two-dimensional perceptrons. From the example, it is clear that the perceptron can shatter 3 points, but it cannot shatter 4 points. Therefore, the break point k for the hypothesis set of two-dimensional perceptrons is $k = 4$.

Verification of $m_{\mathcal{H}}(k) < 2^k$

Next, we need to verify that the growth function $m_{\mathcal{H}}(k)$ is less than 2^k at the break point $k = 4$.

The growth function $m_{\mathcal{H}}(k)$ represents the maximum number of dichotomies that can be implemented by the hypothesis set for k points. According to the formulas in Example 2.2, for small k , the growth function for the perceptron is given by $m_{\mathcal{H}}(k) = k^2 + k + 1$.

For $k = 3$:

$$m_{\mathcal{H}}(3) = 3^2 + 3 + 1 = 9 \quad (\text{but since it can only produce 8 distinct labelings, this matches } 2^3 = 8)$$

For $k = 4$, however, the perceptron cannot shatter 4 points, and hence:

$$m_{\mathcal{H}}(4) < 2^4 = 16.$$

The break point is indeed $k = 4$, and it satisfies the condition that $m_{\mathcal{H}}(k) < 2^k$ for $k = 4$, as required.

3 Exercise 2.2

(a) Verify the bound of Theorem 2.4 in the three cases of Example 2.2:

- (i) Positive rays: $\mathcal{H}$ consists of all hypotheses in one dimension of the form $h(x) = \text{sign}(x - a)$.
- (ii) Positive intervals: $\mathcal{H}$ consists of all hypotheses in one dimension that are positive within some interval and negative elsewhere.
- (iii) Convex sets: $\mathcal{H}$ consists of all hypotheses in two dimensions that are positive inside some convex set and negative elsewhere.

(b) Does there exist a hypothesis set for which $m_{\mathcal{H}}(N) = N + 2^{\lfloor N/2 \rfloor}$, where $\lfloor N/2 \rfloor$ is the largest integer $\leq N/2$?

Solution

(a) Verification of Theorem 2.4 in the three cases of Example 2.2

Theorem 2.4 establishes that if a hypothesis set $\mathcal{H}$ has a break point k , then the growth function $m_{\mathcal{H}}(N)$ is bounded by:

$$m_{\mathcal{H}}(N) \leq \sum_{i=0}^{k-1} \binom{N}{i}$$

This shows that as long as $\mathcal{H}$ has a break point, the growth function will grow polynomially instead of exponentially.

(i) Positive Rays: - Hypothesis Set: $\mathcal{H}$ consists of all hypotheses in one dimension of the form $h(x) = \text{sign}(x - a)$.

- Break Point: The break point for positive rays is $k = 2$, as it can shatter 1 point but cannot shatter 2 points.

- Verification: According to Theorem 2.4, for $k = 2$:

$$m_{\mathcal{H}}(N) \leq \sum_{i=0}^1 \binom{N}{i} = 1 + N$$

This matches the actual growth function for positive rays, which is linear in N beyond 1 point.

(ii) Positive Intervals: - Hypothesis Set: $\mathcal{H}$ consists of all hypotheses in one dimension that are positive within some interval and negative elsewhere.

- Break Point: The break point for positive intervals is $k = 3$, as it can shatter 2 points but not 3 points.

- Verification: For $k = 3$, the bound from Theorem 2.4 is:

$$m_{\mathcal{H}}(N) \leq \sum_{i=0}^2 \binom{N}{i} = 1 + N + \frac{N(N-1)}{2}$$

This is a quadratic function, which corresponds to the quadratic growth of distinct dichotomies for positive intervals.

(iii) Convex Sets: - Hypothesis Set: $\mathcal{H}$ consists of all hypotheses in two dimensions that are positive inside some convex set and negative elsewhere.

- Break Point: The break point for convex sets is $k = 4$, as it can shatter 3 points but cannot shatter 4 points.

- Verification: For $k = 4$, the bound from Theorem 2.4 is:

$$m_{\mathcal{H}}(N) \leq \sum_{i=0}^3 \binom{N}{i} = 1 + N + \frac{N(N-1)}{2} + \frac{N(N-1)(N-2)}{6}$$

This is a cubic function, matching the cubic growth of distinct dichotomies for convex sets.

(b) Hypothesis Set with Special Growth Function

Yes, such a hypothesis set exists. One example is the set of axis-aligned rectangles in two dimensions. For this hypothesis set, the growth function is:

$$m_{\mathcal{H}}(N) = N + 2^{\lfloor N/2 \rfloor}$$

This hypothesis set can generate a number of dichotomies that grows faster than linearly but slower than exponentially, as required by the problem.

4 Exercise 2.3

Compute the VC dimension of $\mathcal{H}$ for the hypothesis sets in parts (i), (ii), and (iii) of Exercise 2.2(a).

Solution

The VC dimension (Vapnik-Chervonenkis dimension) of a hypothesis set $\mathcal{H}$ is the largest number of points that can be shattered by $\mathcal{H}$. If $\mathcal{H}$ can shatter

d points but cannot shatter $d + 1$ points, then the VC dimension is d .

(i) Positive Rays

- Hypothesis Set: This set consists of all hypotheses in one dimension of the form $h(x) = \text{sign}(x - a)$, where the hypothesis changes sign at some point a . - VC Dimension: - The hypothesis set can shatter 1 point because it can classify the point as either positive or negative. - However, it cannot shatter 2 points because it cannot realize all possible labelings of 2 points (there is only one threshold a). - Therefore, the VC dimension of this hypothesis set is 1.

(ii) Positive Intervals

- Hypothesis Set: This set consists of all hypotheses in one dimension that are positive within some interval and negative elsewhere. - VC Dimension: - The hypothesis set can shatter 2 points, as it can realize any labeling of two points (both positive, both negative, or one positive and one negative). - However, it cannot shatter 3 points because it cannot realize all possible labelings of 3 points. For example, it cannot make the middle point negative and the outer points positive. - Therefore, the VC dimension of this hypothesis set is 2.

(iii) Convex Sets

- Hypothesis Set: This set consists of all hypotheses in two dimensions that are positive inside some convex set and negative elsewhere. - VC Dimension: - The hypothesis set can shatter 3 points because a convex set can be constructed that realizes all possible labelings of 3 points. - However, it cannot shatter 4 points because it cannot realize all possible labelings of 4 points in two dimensions. - Therefore, the VC dimension of this hypothesis set is 3.

Conclusion

The VC dimensions of the hypothesis sets are:

- Positive rays: VC dimension = 1
- Positive intervals: VC dimension = 2

- Convex sets: VC dimension = 3

5 Exercise 2.6

A data set has 600 examples. To properly test the performance of the final hypothesis, we set aside a randomly selected subset of 200 examples which are never used in the training phase; these form a test set. A learning model with 1,000 hypotheses is used to select the final hypothesis g based on the 400 training examples. We wish to estimate $E_{\text{out}}(g)$. We have access to two estimates: $E_{\text{in}}(g)$, the in-sample error on the 400 training examples; and $E_{\text{test}}(g)$, the test error on the 200 test examples that were set aside.

- Using a 5% error tolerance ($\delta = 0.05$), which estimate has the higher 'error bar'?
- Is there any reason why we shouldn't reserve even more examples for testing?

Solution

(a) Error Bars for $E_{\text{in}}(g)$ and $E_{\text{test}}(g)$

Using the Hoeffding inequality, we can compute the error bars for both $E_{\text{in}}(g)$ and $E_{\text{test}}(g)$.

The general form of the error bound is:

$$P(|E_{\text{true}}(g) - E_{\text{estimate}}(g)| > \epsilon) \leq 2 \exp(-2\epsilon^2 N)$$

Where N is the number of examples used to calculate the estimate and ϵ is the error bar.

For $E_{\text{in}}(g)$, we have 400 training examples, so the error bar is inversely proportional to $\sqrt{400}$.

For $E_{\text{test}}(g)$, we have 200 test examples, so the error bar is inversely proportional to $\sqrt{200}$.

Since the error bar is proportional to $1/\sqrt{N}$, and $N = 400$ for $E_{\text{in}}(g)$ and $N = 200$ for $E_{\text{test}}(g)$, the error bar for $E_{\text{test}}(g)$ will be larger. Specifically:

$$\frac{\text{Error bar for } E_{\text{test}}(g)}{\text{Error bar for } E_{\text{in}}(g)} = \sqrt{\frac{400}{200}} = \sqrt{2}$$

The test error $E_{\text{test}}(g)$ has a higher error bar.

(b) Reserving More Examples for Testing

Reserving more examples for testing comes at the expense of having fewer examples for training. This leads to a trade-off:

- If we reserve more examples for testing, we can reduce the error bar for $E_{\text{test}}(g)$, but fewer training examples will lead to a less accurate hypothesis g since the hypothesis will be trained on less data.
- If we use fewer examples for testing and more for training, we get a better hypothesis g due to more training data, but the test error estimate will be less reliable due to fewer test examples.

There is a balance between the size of the training set and the test set. Setting aside too many examples for testing could hurt the accuracy of the hypothesis learned during training.

Conclusion

(a) The error bar for $E_{\text{test}}(g)$ is larger than the error bar for $E_{\text{in}}(g)$, since the test set is smaller.

(b) We should not reserve too many examples for testing because it would leave fewer examples for training, potentially leading to a worse hypothesis g .

6 Problem 1.11

The matrix which tabulates the cost of various errors for the CIA and Supermarket applications in Example 1.1 is called a risk or loss matrix.

For the two risk matrices in Example 1.1, explicitly write down the in-sample error E_{in} that one should minimize to obtain g . This in-sample error should weight the different types of errors based on the risk matrix. [Hint: Consider $y_n = +1$ and $y_n = -1$ separately.]

Solution

In this problem, we are given risk or loss matrices that define the cost of errors in two different applications: the CIA and the Supermarket. The goal is to compute the in-sample error E_{in} that should be minimized, taking into account the different costs of false positives and false negatives based on these matrices.

CIA Application

For the CIA application, the risk matrix gives a much higher penalty for a false positive (accepting an unauthorized person) than for a false negative (rejecting an authorized person). The matrix is:

$$\begin{pmatrix} 0 & 1000 \\ 10 & 0 \end{pmatrix}$$

The cost for a false positive, when $y_n = -1$ but $g(x_n) = +1$, is 1000. The cost for a false negative, when $y_n = +1$ but $g(x_n) = -1$, is 10.

The weighted in-sample error E_{in} for the CIA application can be written as:

$$E_{\text{in}} = \frac{1}{N} \sum_{n=1}^N [1000 \cdot \mathbb{I}(y_n = -1 \text{ and } g(x_n) = +1) + 10 \cdot \mathbb{I}(y_n = +1 \text{ and } g(x_n) = -1)]$$

Where $\mathbb{I}(\cdot)$ is the indicator function that is 1 when its argument is true and 0 otherwise.

Supermarket Application

For the Supermarket application, the costs are more balanced. The risk matrix is:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

In this case, both false positives (identifying a non-member as a member) and false negatives (failing to identify a member) have equal costs of 1.

Therefore, the in-sample error E_{in} for the Supermarket application is simply:

$$E_{\text{in}} = \frac{1}{N} \sum_{n=1}^N [\mathcal{K}(y_n = -1 \text{ and } g(x_n) = +1) + \mathcal{K}(y_n = +1 \text{ and } g(x_n) = -1)]$$

This gives equal weight to both types of errors.

Conclusion

The in-sample error that should be minimized depends on the specific costs of false positives and false negatives as defined by the risk matrix. For the CIA application, false positives are much more costly than false negatives, so the in-sample error heavily weights false positives. For the Supermarket application, the costs are balanced, and both types of errors are weighted equally in the in-sample error.

7 Problem 1.12

This problem investigates how changing the error measure can change the result of the learning process. You have N data points $y_1 \leq \dots \leq y_N$ and wish to estimate a 'representative' value.

- (a) If your algorithm is to find the hypothesis h that minimizes the in-sample sum of squared deviations:

$$E_{\text{in}}(h) = \sum_{n=1}^N (h - y_n)^2,$$

show that your estimate will be the in-sample mean:

$$h_{\text{mean}} = \frac{1}{N} \sum_{n=1}^N y_n.$$

- (b) If your algorithm is to find the hypothesis h that minimizes the in-sample sum of absolute deviations:

$$E_{\text{in}}(h) = \sum_{n=1}^N |h - y_n|,$$

show that your estimate will be the in-sample median h_{med} , which is any value for which half the data points are at most h_{med} and half the data points are at least h_{med} .

- (c) Suppose y_N is perturbed to $y_N + \epsilon$, where $\epsilon \rightarrow \infty$. So, the single data point y_N becomes an outlier. What happens to your two estimators h_{mean} and h_{med} ?

Solution

(a) Minimizing the Sum of Squared Deviations

We want to minimize the following in-sample error:

$$E_{\text{in}}(h) = \sum_{n=1}^N (h - y_n)^2.$$

To find the hypothesis h that minimizes this error, we take the derivative of $E_{\text{in}}(h)$ with respect to h and set it equal to zero:

$$\frac{d}{dh} \left(\sum_{n=1}^N (h - y_n)^2 \right) = \sum_{n=1}^N 2(h - y_n) = 0.$$

This simplifies to:

$$2Nh - 2 \sum_{n=1}^N y_n = 0,$$

$$h = \frac{1}{N} \sum_{n=1}^N y_n.$$

The hypothesis h that minimizes the sum of squared deviations is the in-sample mean:

$$h_{\text{mean}} = \frac{1}{N} \sum_{n=1}^N y_n.$$

(b) Minimizing the Sum of Absolute Deviations

Now, we want to minimize the sum of absolute deviations:

$$E_{\text{in}}(h) = \sum_{n=1}^N |h - y_n|.$$

To minimize this, we need to find h such that the total absolute distance from h to the data points is minimized. This happens when h is the median of the data points.

At the median h_{med} , half of the data points are less than or equal to h_{med} , and half of the data points are greater than or equal to h_{med} . This balances the total absolute deviations on both sides, minimizing the sum.

The hypothesis h that minimizes the sum of absolute deviations is the in-sample median h_{med} .

(c) Effect of Perturbing y_N on h_{mean} and h_{med}

Suppose y_N is perturbed to $y_N + \epsilon$, where $\epsilon \rightarrow \infty$. This means that the last data point becomes an outlier.

- Effect on h_{mean} : The in-sample mean h_{mean} is highly sensitive to outliers because it depends on the sum of all data points. As $y_N \rightarrow \infty$, the mean h_{mean} will increase significantly, as it will be pulled towards the large outlier.
- Effect on h_{med} : The in-sample median h_{med} is robust to outliers. Since the median only depends on the middle data points, a change in y_N (the largest data point) will not affect the median as long as y_N is still on the extreme end. Therefore, h_{med} will remain relatively unaffected by the perturbation.

Conclusion

- (a) The hypothesis that minimizes the sum of squared deviations is the in-sample mean $h_{\text{mean}} = \frac{1}{N} \sum_{n=1}^N y_n$.
- (b) The hypothesis that minimizes the sum of absolute deviations is the in-sample median h_{med} .
- (c) When y_N becomes an outlier, h_{mean} is heavily affected, while h_{med} remains stable.